# HALO

## HIGH ACUITY LOW OCCURRENCE EMERGENCY PROCEDURES

**⚠ WARNING: ADULT PATIENTS ONLY**

STRICTLY FOR USE IN PATIENTS AGED 16 AND OVER. DO NOT USE FOR PAEDIATRIC EMERGENCIES.

**2026 MASTER EDITION**

Version 2.9 (Sep 2026)

Based on DAS, RCEM, FPHC and Resuscitation Council UK Guidelines. Review due Sep 2027.

v2.9 changes: thoracotomy (EMJ 2026), subclavian (Safe Vascular Access 2025, MSD; large-bore introducer), breech (RCOG 20b), escharotomy (LSEBN 2026, MSD), pacing (RCUK 2025), canthotomy (RCEM, MSD), chest drain LA max dose, eFONA corrections. All procedure diagrams redrawn for this handbook (Oct 2026; base artwork AI-generated, labels added and checked by hand).

⚠️ **STRICTLY FOR PATIENTS AGED 16 AND OVER**

# HOW TO USE THIS GUIDE

## 1. The Two-Page System

Every procedure consists of two sequential pages:

**Page A: The Walkthrough (Preparation)**
Use this during the "Kit Dump" phase. It includes specific equipment checklists and detailed setup instructions.

**Page B: The Flowchart (Action)**
Use this during the procedure itself. Designed for high-stress cognitive processing.

## 2. Design Principles

**Standard Steps (Grey):** Routine actions and setups.
**Critical Steps (Red):** Dangerous actions (incisions) or stops.
**Plan B (Red Box):** Fallback plans if the primary method fails.

## 3. Human Factors & Safety

**Allocate a Reader:** Hand this book to a specific team member. Instruct them to read the steps aloud to reduce cognitive load for the operator.
**Pre-procedure Pause:** Pause for 5 seconds before making any incision. Verbally confirm equipment, landmarks, and check the patient for severe allergies.

## 4. Glossary of Abbreviations

| | | | |
|---|---|---|---|
| **DAS** | Difficult Airway Society | **RCEM** | Royal College of Emergency Medicine |
| **FPHC** | Faculty of Pre-Hospital Care | **RCOG** | Royal College of Obstetricians & Gynaecologists |
| **LSEBN** | London & South East Burns Network | **BTS** | British Thoracic Society |
| **ESC** | European Society of Cardiology | **AAGBI** | Association of Anaesthetists |
| **ALS** | Advanced Life Support | **NLS** | Neonatal Life Support |
| **ROSC** | Return of Spontaneous Circulation | **TXA** | Tranexamic Acid |
| **IOP** | Intraocular Pressure | **CICO** | Can't Intubate Can't Oxygenate |
| **CTM** | Cricothyroid Membrane | | |

⚠️ **CLINICAL DISCLAIMER**
**These guides are cognitive aids for trained medical professionals. They do not replace clinical judgement. Guidelines are subject to change.**

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🧰 = Item located in HALO Trolley


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# RESUSCITATIVE HYSTEROTOMY

**Page 1: Walkthrough (Preparation)**

🧰 **HALO TROLLEY**

## INITIAL ACTIONS AND INDICATIONS CHECK

- Call 2222 for "Maternal Cardiac Arrest" and "Neonatal Emergency".
- Declare Obstetric Major Haemorrhage Protocol.
- **Assign Roles:**

1. Person to manage Major Haemorrhage (blood/transfusion).
2. Person to lead ALS (Defib/Drugs).
3. Assign a team just to manage the baby.

- **Start ALS + Obstetric Modifications:**
- Manually displace uterus to patient's LEFT.
- Intubate early (smaller ETT e.g., 6.0/7.0).
- IV access x2 (Above Diaphragm).

## EQUIPMENT & SETUP

- ☐ Chlorhexidine (Splash).
- ☐ **Sterile Drape / Gown Pack.**
- ☐ **Scalpel:** Size 10 or 20.
- ☐ **Tuff Cut Scissors.**
- ☐ **Suction x2** (Wide bore/Yankauer).
- ☐ Large Swabs (for packing).
- ☐ Stapler / Clingfilm / Large Dressings.
- ☐ **Umbilical Cord Clamps.**
- ☐ **Artery Forceps** (x2).
- ☐ **Scissors or Extra Scalpel** (for cord).

## 📍 STORAGE LOCATION

**HYSTEROTOMY KIT IS IN THE HALO TROLLEY. NEONATAL RESUSCITATION EQUIPMENT IS KEPT IN PAEDIATRIC RESUS.**

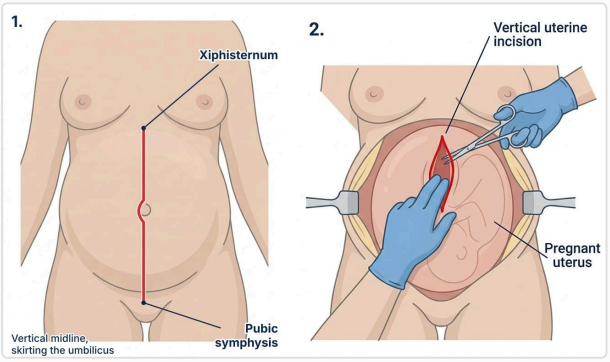
## POST-PROCEDURE PLAN

**If ROSC Achieved:**

- Mass closure of abdomen with large dressings, clingfilm or staples.
- Immediate Transfer: Emergent Theatre with Obstetrics.
- Continue ALS / Stabilisation during transfer.

## PROCEDURE WALKTHROUGH

1. **Clean:** Rapid splash with CHLORHEXIDINE.
2. **Skin Incision:** Vertical incision from Xiphisternum to Pubic Symphysis. Cut deep through skin and fat with SCALPEL.
3. **Peritoneum:** Bluntly enter peritoneum with fingers (tear it open) to avoid bowel injury.
4. **Uterus Exposure:** Retract abdominal wall laterally.
5. **Uterine Incision:** Vertical incision in the upper segment of uterus. Use SCALPEL to start, then TUFF CUT SCISSORS.
6. **Protect Baby:** Insert two fingers to lift uterine wall away from fetal head. Extend cut with TUFF CUT SCISSORS.
7. **Delivery:** Scoop head and shoulders. Do not pull by a single limb. Rotate if necessary.
8. **Placenta:** One gentle tug with smooth traction. If not easy, leave in situ. Pack uterus with SWABS.




*Vertical Midline Incision: Speed is priority*

# RESUSCITATIVE HYSTEROTOMY

**Page 2: Action (Perimortem C-Section)**

🧰 **HALO TROLLEY**

- **GOAL: DELIVERY in < 5 MINS (Timer starts at the exact moment of maternal cardiac arrest).**
- **Indication: Maternal Arrest. Uterus greater than or equal to Umbilicus.**
- **Call: 2222 Maternal Cardiac Arrest.**

## 1 ABDOMINAL INCISION

- Pause for 5 seconds. Confirm landmarks.
- **Vertical Midline:** Xiphisternum to Pubic Symphysis.
- Incise deep through skin/fat. Separate rectus muscles bluntly.
- Ignore bleeding; there is no cardiac output.

## 2 UTERINE INCISION

- Enter peritoneum bluntly with fingers.
- **Vertical Incision:** Incise uterus (Avoids bladder/arteries).
- Insert fingers to protect baby's face. Extend with TUFF CUT SCISSORS.

## 3 DELIVERY

- **Scoop head and shoulders.** Do not pull on a single limb.
- Rotate if necessary to free shoulders.
- Clamp and Cut Cord. Handover to NLS Team.

## 4 POST-DELIVERY & CLOSURE

- **Placenta:** One gentle tug. If stuck, leave in situ. One hand sweep if bleeding.
- Check allergies. **Drugs:** Oxytocin `5iu` slow IV (never rapid bolus), Antibiotics, TXA `1g`. FETCH DRUG
- **Closure:** Pack uterus. Close abdomen with dressings, clingfilm or staples.
- **Transfer:** Emergent Theatre (Obstetrics) if ROSC.



*Feedback/Updates: jake.turner@uhb.nhs.uk*

Reference: Resuscitation Council UK (2025 Special Circumstances).
Open Guideline

# EMERGENCY BREECH

Page 1: Walkthrough (Preparation)

## INITIAL ACTIONS AND INDICATIONS CHECK

- Call 2222 for "Obstetric Emergency" and "Neonatal Emergency".
- **Position:** Semi-recumbent, buttocks to edge of bed. All-fours only if operator experienced; move to semi-recumbent for manoeuvres.
- **Team:** Obstetric Reg, Midwife, Neonatal Team, Anaesthetics.
- **HANDS OFF:** Do NOT touch baby until umbilicus is visible.
- **Gravity:** Allow baby to hang. Avoid traction. Minimise touching (stimulation extends arms/head).
- **Intervene if:** >5 min buttocks→head, >3 min umbilicus→head, poor tone/colour, extended arms or neck.
- Grasp by the **bony pelvis only**. **Never hyperextend the neck.**

## SPECIFIC EQUIPMENT

- [ ] Chlorhexidine.
- [ ] Sterile Drape.
- [ ] Neonatal Resuscitation Kit.
- [ ] Oxytocin (for post-delivery). FETCH DRUG
- [ ] GTN Spray (Uterine relaxation). FETCH DRUG
- [ ] Episiotomy Scissors (selective, not routine).
- [ ] Resuscitaire (Heater On).

## 📍 STORAGE LOCATION

**NEONATAL RESUSCITATION EQUIPMENT IS KEPT IN PAEDIATRIC RESUS.**

## POST-PROCEDURE PLAN

- Inspect for perineal tears.
- Paediatric review for baby.
- **Transfer:** Obstetric Unit / Theatre for repair.

## MANOEUVRES (IF DESCENT STOPS)

1. **Legs Stuck (Pinard's):**
   - Insert fingers behind fetal knee (popliteal fossa).
   - Press to **abduct** thigh (move away from midline).
   - Leg will flex. Sweep it out.
2. **Arms:** Scapula visible → finger in elbow, flex arm down across the chest.
   **Nuchal / stuck arm (Lovset's):**
   - Hold bony pelvis (thumbs on sacrum).
   - Rotate baby 180 degrees (Keep back up/anterior).
   - Posterior arm sweeps down across the chest. Deliver arm.
   - Rotate back 180 degrees to deliver second arm.
3. **Head Stuck (Mauriceau-Smellie-Veit):**
   - Assistant: Suprapubic pressure (flexes head).
   - Operator: Rest baby on forearm ("Flying horse").
   - Fingers on cheekbones (malar). Flex head. Pull gently.
   - Do NOT use Burns-Marshall (overextends neck).

# EMERGENCY BREECH

**Page 2: Action (Vaginal Manoeuvres)**

- **THE GOLDEN RULE: HANDS OFF.**
- **Allow breech to hang by gravity.**
- **Keep room warm. Note time of buttock delivery. Intervene: >5 min buttocks→head / >3 min umbilicus→head, poor tone, extended arms/neck.**
- **Grasp bony pelvis only. Never hyperextend the neck.**

## 1 LEGS STUCK (PINARD'S)

- **Indication:** Frank Breech (legs extended).
- Finger in popliteal fossa.
- Abduct thigh laterally. Knee flexes. Sweep out.

## 2 ARMS (LOVSET'S IF NUCHAL/STUCK)

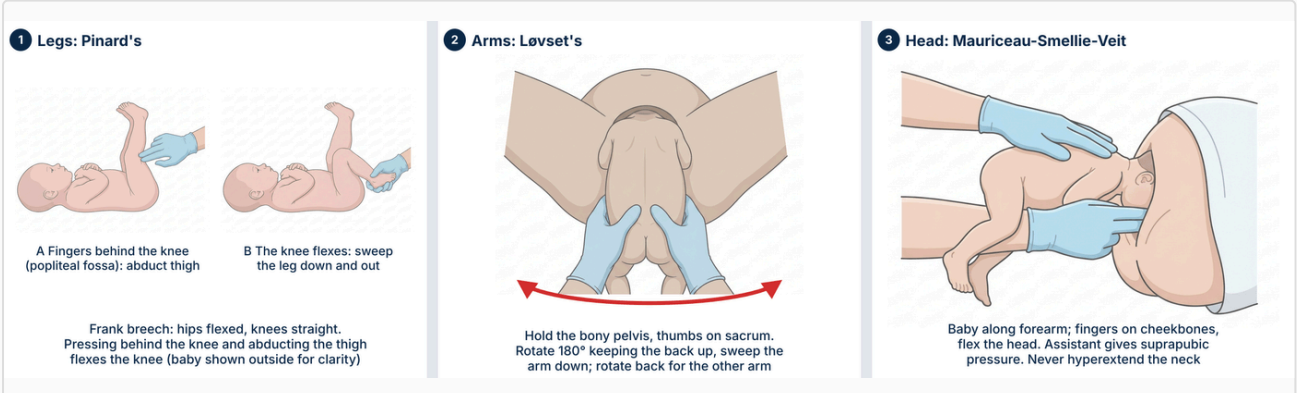
- **Scapula visible:** finger in elbow, flex arm down across chest.
- **Nuchal/stuck:** hold bony pelvis. Rotate 180 degrees. Arm sweeps down across chest.
- Rotate back 180 degrees for second arm.

## 3 HEAD STUCK (MSV)

- **Mauriceau-Smellie-Veit:** Ensure clear airway.
- Check for umbilical cord around neck. Slip it over the head if present.
- Baby on forearm. Fingers on cheekbones. Flex head.
- Tip: Assistant applies Suprapubic Pressure. Do NOT use Burns-Marshall.

## 4 POST-PROCEDURE

- Inspect for perineal tears.
- Paediatric review for baby (Check hips/limbs).
- Active management of third stage (Oxytocin `10iu` IM).




*Pinard's (legs), Løvset's (arms), Mauriceau-Smellie-Veit (head)*

*Feedback/Updates: jake.turner@uhb.nhs.uk*

Reference: RCOG Green-top Guideline 20b (2017).
Open Guideline

# RESUSCITATIVE THORACOTOMY

**Page 1: Walkthrough (Preparation)**

🧰 **HALO TROLLEY**

## INITIAL ACTIONS AND INDICATIONS CHECK

• Call 2222 "Trauma Cardiac Arrest". Activate Major Haemorrhage Protocol. O-negative blood.
• Consider WMAS MERIT / HEMS: patient will need transfer to QE.
• **ERC 2025:** ≤15 min from loss of vital signs.
• **Penetrating:** Arrest <15 min. **Blunt:** <10 min (highly selective).
• **Signs of Life:** Pupils reacting, gasping, ECG activity, cardiac motion on US.
• **Best results before output is fully lost.** Tamponade can survive ~15 min; exsanguination rarely >5 min.

## TEAM & SET-UP

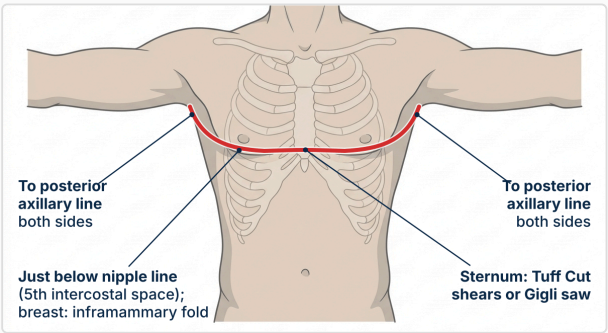
• **Minimum 2:** OPERATOR (surgical field only) + ASSISTANT (ventilation timing, instruments, **sharps + swab count**, scene).
• **Arms out (crucifix)** before cutting: arm by side shortens incision + hampers aortic compression.
• **Ventilation:** SGA is fine. PAUSE for pleural entry, pericardium, heart inspection, hilar control. Resume small tidal volumes once massage/output.
• Gown/gloves for sharps risk (saw, shears, rib edges). Good lighting.

## EQUIPMENT TRAY

- ☐ Chlorhexidine, sterile drapes/gowns, **sterile skin marker**.
- ☐ **Scalpel** size 22 + **curved scissors**.
- ☐ **Tuff Cut shears** + **Gigli saw** (handles + wire).
- ☐ **Finochietto retractor** (check assembled correctly).
- ☐ Toothed forceps (Bonney's), artery forceps (Spencer Wells).
- ☐ **Non-crushing vascular clamps** (Satinsky/DeBakey).
- ☐ 2-0 Prolene/silk on **curved non-cutting needle**, skin stapler.
- ☐ **Counted swabs**, NG tube, internal paddles.
- ☐ Foley 20F (last resort), large dressings, cling film.

## CLAMSHELL TECHNIQUE

1. **MARK:** Just below nipple line (5th ICS). Breast tissue: lift breast, use **inframammary fold**.
2. **SKIN:** One arc, skin + subcut, towards **POSTERIOR axillary line** both sides (spare latissimus dorsi). Join thoracostomies if correctly sited.
3. **PLEURA:** CURVED SCISSORS along **upper border of rib**. Never enter pleura with scalpel. Different ICS each side? Extend one across sternum to match.
4. **STERNUM:** TUFF CUT SHEARS first. Gigli saw: assistant holds Spencer Wells flat across sternum to stop saw springing up.
5. **OPEN:** FINOCHIETTO, handle down, crank open. Release sterno-pericardial ligament.
6. **PERICARDIUM:** Lift with forceps, nick centre, ONE continuous vertical cut head-to-toe, ANTERIOR to phrenic nerve. Evacuate clot.
7. **DELIVER HEART:** Inspect all surfaces for wounds.
8. **THEN:** Cardiac wounds → lung → aorta → massage (see page 2).




*Clamshell: towards posterior axillary line, both sides*

## 📍 STORAGE LOCATION

**THORACOTOMY KIT (FINOCHIETTO RETRACTOR, GIGLI SAW) AND INTERNAL PADDLES ARE IN THE THORACOTOMY DRAWER OF THE HALO TROLLEY.**

# RESUSCITATIVE THORACOTOMY

**Page 2: Action ("The Clamshell")**

🧰 **HALO TROLLEY**

- **≤15 min from loss of vital signs** (ERC 2025); blunt <10 min. 2222 + MHP.

## 1 OPEN THE CHEST (PAUSE VENTILATION)

- **Scalpel** skin → **scissors** intercostals (upper rib border) → **Tuff Cut** sternum → Finochietto.
- **Pericardium:** one vertical cut, anterior to phrenic nerve. Evacuate clot. Deliver heart.

## 2 THEN, IN ORDER (FIX BLEEDING AS FOUND)

- **Heart wound: finger / direct pressure first.** Suture: curved non-cutting needle, mattress suture. Don't overtighten (cuts through as heart fills). **Foley = last resort** (enlarges defect).
- ⚠️ **DO NOT OCCLUDE CORONARY ARTERIES.** Near one: suture DEEP to it, or hold pressure.
- **Lung bleeding:** counted swabs → non-crushing clamp → hilar control. Lung twist = last resort.
- **Aorta:** release inferior pulmonary ligament. Palm up to spine: first tube = aorta, then oesophagus (NG helps). **Compress with palm/fist against spine, near diaphragm. No pinching. Clamp only if trained.**
- **Internal massage: flat hands**, both palms, no fingertips. `80–100/min`, allow filling. Empty heart = give volume. Sole job; rotate early.

## 3 VF: FIX FLOW FIRST, THEN SHOCK

- Traumatic VF = poor coronary flow: **fix bleeding, aorta, volume, massage first.**
- **Internal paddles:** `10–20 J`, up to `40 J` if first shock fails.
- **No paddles:** partly close chest. Pads **anteroposterior**; anterior pad must NOT cross incision. `200J`. No one touching.

## 4 ROSC

- Clamp bleeding internal mammary arteries. **Partly close; leave retractor in, loosened**, for re-entry.
- **Keep warm** (cling film, blankets). Sedate + paralyse. Theatre with cardiothoracics.
- Ongoing aortic compression: **45 s on / 15 s off**. Isolated tamponade: avoid over-resuscitation.

**🛑 STOP CRITERIA**

Injuries incompatible with life, or no ROSC despite tamponade release, adequate volume + aortic occlusion ($ETCO_2$ and cardiac filling help decide).
Survival ~2% in exsanguination; none if tamponade + exsanguination coexist. Beware bias to continue.
Consensus decision. Debrief the team.

*Feedback/Updates: jake.turner@uhb.nhs.uk*

Reference: Chesters A et al. Emergency thoracotomy: how to do it in 2026. Emerg Med J 2026 (doi:10.1136/emermed-2026-216110) | South East Scotland Major Trauma Guidelines - Resuscitative Thoracotomy | FPHC Traumatic Cardiac Arrest Position Statement (2024).

Open Guideline

# ESCHAROTOMY

**Page 1: Walkthrough (Preparation)**

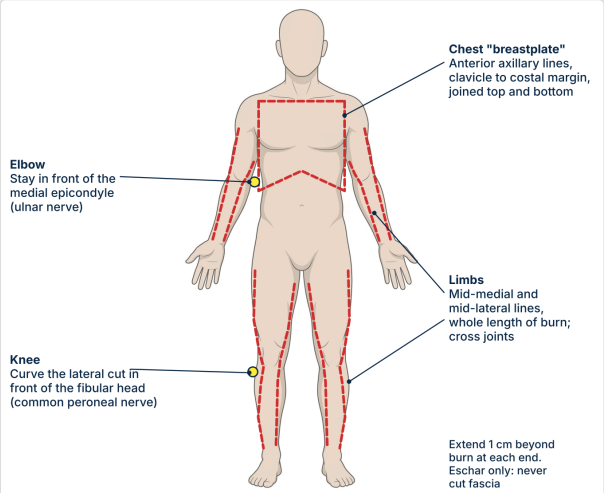
🧰 **HALO TROLLEY**

## INITIAL ACTIONS AND INDICATIONS CHECK

• 2222 trauma team. **Discuss with burns service (QE) BEFORE cutting**, unless ventilation is failing now (LSEBN).
• **Circumferential** full thickness / deep dermal burns (painless, leathery, white/charred).
• **Chest/neck/torso:** Ventilation difficulty. High peak pressures (> 30). Hypoxia.
• **Limbs:** Absent Doppler flow or distal $SpO_2$ <95% (without systemic hypotension/hypoxia). Loss of pulses is late. Skin temperature is unreliable.

## EQUIPMENT

- [ ] Chlorhexidine.
- [ ] **Sterile Drape / Gown Pack.**
- [ ] **Scalpel** (Size 22).
- [ ] **Sterile Skin Marker** (Draw lines first).
- [ ] **Artery Forceps** + Gauze (pressure).
- [ ] Lidocaine 1%, 10ml syringe, 25G + 21G needles. FETCH DRUG
- [ ] Cling Film + Loose Bandages (Crepe).




*Mark lines before cutting*

## INCISION TECHNIQUE

1. **MARK:** Draw lines with MARKER. Extend **1cm beyond burn** at each end; cross joints.
2. **ANALGESIA:** Full thickness is painless; LA/opioid for deep partial-thickness edges.
3. **DEPTH:** Cut through eschar down to subcutaneous fat with SCALPEL.
   **STOP:** Never cut fascia or muscle.
4. **RELEASE:** Edges must gape widely; run finger along cut. **Bleeding should be minimal** at the right depth: press; artery forceps for a discrete vessel. **Brisk bleeding = too deep.**
5. **CHEST ("Breastplate"):** Anterior axillary lines, clavicle to costal margin (avoid breast tissue); join across top and bottom (subcostal chevron).
6. **NECK:** Lateral and posterior (avoid carotid/jugular).
7. **UPPER LIMB:** Mid-lateral + mid-medial. **Elbow:** ANTERIOR to medial epicondyle (ulnar nerve). **Wrist:** avoid radial nerve. **Hand:** dorsal over metacarpals; extend to thenar/hypothenar.
8. **LOWER LIMB:** Mid-lateral + mid-medial. **Knee:** curve lateral cut around fibular neck (common peroneal). **Ankle:** avoid posterior tibial vessels (medial). **Foot:** to great toe medially, little toe laterally.

## POST-PROCEDURE PLAN

• Reassess: Doppler / distal $SpO_2$. Not improved → recheck depth; never cut fascia.
• Cover: loose longitudinal cling film strips or dressing agreed with burns. Loose bandage.
• Elevate limbs. **Keep warm.**
• **Transfer:** Specialist Burns Centre / ICU.

# ESCHAROTOMY

**Page 2: Action (Burns Release)**

🧰 **HALO TROLLEY**

- **Call:** 2222 Trauma / Medical Emergency.
- **Discuss with burns (QE) before cutting** unless ventilation is failing now.
- **Chest:** Ventilatory compromise. **Limbs:** Absent Doppler / distal $SpO_2$ <95%.
- Analgesia/sedation — surrounding partial-thickness burns are painful.

## 1 CHEST ("BREASTPLATE")

- Pause for 5 seconds. Confirm landmarks.
- **Vertical:** Bilateral anterior axillary lines, clavicle to costal margin. Avoid breast tissue.
- **Transverse:** Connect tops (clavicle) and bottoms (costal margin / subcostal chevron).
- Result: Chest wall must spring open. **Neck:** lateral + posterior cuts.

## 2 UPPER LIMBS

- **Lines:** Mid-Lateral and Mid-Medial, 1cm beyond burn, across joints.
- **Hand:** Dorsal over metacarpals; extend to thenar/hypothenar.
- Ulnar Nerve: Cut ANTERIOR to Medial Epicondyle. Wrist: avoid radial nerve.

## 3 LOWER LIMBS

- **Lines:** Mid-Lateral and Mid-Medial.
- **Foot:** to great toe medially, little toe laterally.
- Peroneal Nerve: Curve around the fibular neck. Ankle: avoid posterior tibial vessels.

## 4 POST-PROCEDURE

- Reassess Doppler / distal $SpO_2$. Not improved → recheck depth; do NOT cut fascia — discuss fasciotomy with burns/plastics.
- Bleeding should be minimal: press; clamp a discrete vessel. Brisk bleeding = too deep.
- Cover: loose longitudinal cling film strips (or burns-agreed dressing). Loose bandage.
- Elevate limbs. Keep warm.
- **Transfer:** Specialist Burns Centre / ICU.



*Feedback/Updates: jake.turner@uhb.nhs.uk*

Reference: LSEBN Initial Management of Severe Burns (2026) | MSD Manual: Burn Escharotomy (2025).
Open Guideline

# CHEST DRAIN (OPEN)

**Page 1: Walkthrough (Preparation)**

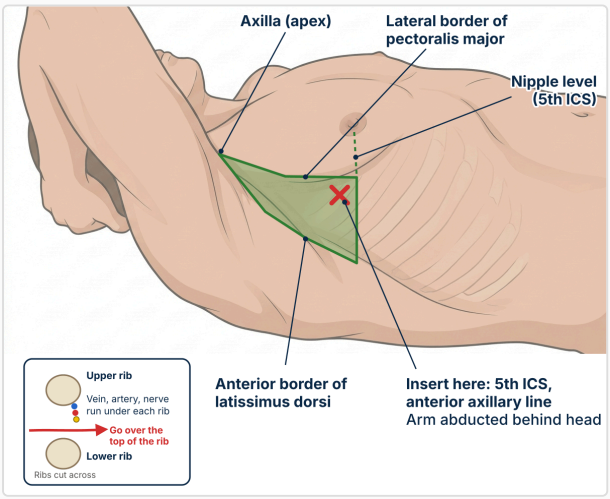
🧰 **HALO TROLLEY**

## INITIAL ACTIONS AND INDICATIONS CHECK

- **Trauma:** Haemothorax, Pneumothorax (ventilated/unstable).
- **Medical:** Empyema, Massive Effusion.
- **Tension Pneumothorax:** Decompress with needle/finger FIRST if arresting.
- **Contraindications:** Coagulopathy (Relative), Diaphragmatic Hernia.

## EQUIPMENT

- ☐ Chlorhexidine + Sterile Drape / Gown Pack.
- ☐ Lidocaine **1%** (`20ml` = 200mg). FETCH DRUG **Max 3mg/kg plain (7mg/kg with adrenaline) — TOTAL across all procedures.**
- ☐ Scalpel (Size 10).
- ☐ **Spencer Wells Forceps** (Straight).
- ☐ Curved Kelly Forceps (Alternative).
- ☐ **Chest Tube:** 28-32F (Trauma); 12-14F Seldinger kit (Air).
- ☐ Silk Suture (size 0).
- ☐ Underwater Seal (Primed).
- ☐ Dressings (Tegaderm/Mepore).




*Triangle of Safety: Lat Dorsi, Pec Major, Axilla, Nipple*

## OPEN TECHNIQUE

1. **POSITION:** Supine. Arm abducted behind head. Secure arm.
2. **LANDMARK:** Safety Triangle. 5th Intercostal Space, Anterior Axillary Line.
3. **LOCAL:** Check allergies. Infiltrate skin, muscle, and pleura. Aspirate fluid/air to confirm depth.
4. **INCISION:** 3 to 4cm Transverse cut parallel to ribs with SCALPEL.
5. **DISSECT:** Use Straight SPENCER WELLS FORCEPS. Push closed, open to spread. Tunnel over the **UPPER BORDER** of the rib below (Avoid intercostal vessels).
6. **BREACH:** Push forceps firmly through pleura ("POP"). Open jaws wide to dilate track.
7. **SWEEP:** Insert finger. Rotate 360 degrees. Feel lung. Break adhesions. **Finger Sweep** mandatory. Confirm you are in chest, not abdomen.
8. **INSERT:** Clamp CHEST TUBE tip with forceps. Guide into chest. Aim Apical (Air) or Basal (Blood). Connect seal. Secure with SILK SUTURE.

## POST-PROCEDURE PLAN

- Connect to underwater seal.
- Secure with sutures/dressings.
- **Check Position:** Request CXR to confirm placement.
- **Monitor Output:** Massive output (> 1500ml stat or > 200ml/hr for 2–4 hrs) requires immediate cardiothoracic involvement.

# CHEST DRAIN (OPEN)

**Page 2: Action (Trauma / Haemothorax)**

🧰 **HALO TROLLEY**

## 1 SAFETY TRIANGLE

- Lat Dorsi (Posterior). Pec Major (Anterior). Axilla (Superior).
- Target: **5th ICS**, Anterior Axillary Line.

## 2 DISSECTION

- Pause for 5 seconds. Confirm landmarks.
- Check allergies. Give LA (max 3mg/kg plain, 7mg/kg with adrenaline — total dose). 3cm Transverse incision.
- Blunt dissection with Spencer Wells Forceps.
- Stay on Upper Border of Lower Rib (avoid vessels).

## 3 PLEURAL ENTRY

- Breach pleura with forceps ("Pop"). Dilate track.
- **Finger Sweep** mandatory (Verify cavity).
- Insert Tube. Look for fogging/swinging.

🛑 **PLAN B: CANNOT INSERT?**

- Enlarge incision. Insert Finger again.
- Use Bougie to guide tube.
- Decompress with Finger Thoracostomy if arresting.

## 4 SECURING & POST-PROCEDURE

- Connect to underwater seal.
- **Suture:** Mattress or Stay suture (size 0 Silk). Tape securely.
- **Post-Procedure:** Request CXR to confirm position. Monitor drain output. Massive output (> 1500ml stat or > 200ml/hr for 2–4 hrs) requires immediate cardiothoracic involvement and possible thoracotomy.



*Feedback/Updates: jake.turner@uhb.nhs.uk*

Reference: RCEM Learning - Chest Drain Insertion in Adult Trauma | BTS Pleural Disease Guideline (2023).
Open Guideline

# PERICARDIOCENTESIS

Page 1: Walkthrough (Preparation)

🧰 HALO TROLLEY

## INITIAL ACTIONS AND INDICATIONS CHECK

• Call 2222 "Cardiac Arrest" or "Medical Emergency". Call Cardiology urgently.
• **Indication:** CLINICAL tamponade — dyspnoea, raised JVP, hypotension, tachycardia, pulsus paradoxus. **Emergency drainage if haemodynamically unstable.** Echo signs alone are NOT an indication.
• **POCUS:** Effusion + RA/RV diastolic collapse.
• **Contraindications:** None absolute in shock. Aortic dissection / post-MI free-wall rupture → surgery (if unavailable or too unstable: drain small volumes to SBP ~90 as bridge). Relative: coagulopathy, anticoagulants, platelets <50.
• **Trauma:** Thoracotomy is superior. Failed aspiration does NOT exclude haemopericardium (clot; 20–40% false negative).

## EQUIPMENT

- [ ] Chlorhexidine + Sterile Drape / Gown Pack.
- [ ] Ultrasound (Sterile Probe Cover + Gel).
- [ ] Cannula: 16G (grey) or 14G (orange).
- [ ] Central line kit (quad lumen, 20 cm): J-wire, dilator, catheter = drain.
- [ ] 20ml Syringe + 3-Way Taps ×2.
- [ ] Agitated Saline (5ml). FETCH SALINE

## ECHO-GUIDED APPROACH

**Position:** Semi-reclining ~30°, slightly rotated to LEFT.
**Entry site:** Where the pericardial space is closest to the probe and fluid is deepest, with no organs in the way (often para-apical).

1. **Apical:** 1–2cm lateral to apex beat, 5th–7th ICS. Over upper border of rib, along echo path. Risk: LV puncture, pneumothorax.
2. **Sub-Xiphoid:** Between xiphisternum and left costal margin. Under ribs, lower to 15–30°, aim LEFT shoulder. Too steep → peritoneum; too medial → RA.
3. **Parasternal:** 5th left ICS close to sternal margin, perpendicular. >1cm lateral = internal thoracic vessels. Risk pneumothorax.

## POST-PROCEDURE PLAN

• CXR (exclude pneumothorax). Urgent Cardiology/CT Surgery.
• Leave central line as drain: aspirate every 4–6h; remove when <25–30ml/24h (usually 24–72h).

# PERICARDIOCENTESIS

**Page 2: Action (Tamponade Decompression)**

🧰 **HALO TROLLEY**

- **Clinical tamponade + haemodynamic compromise = DRAIN NOW. Echo signs alone are not an indication.**
- **In Traumatic Arrest use Thoracotomy. Call: 2222 Cardiac Arrest.**

## 1 TEMPORISE

- SBP <100 + hypovolaemic: `250–500ml` saline (larger volumes can worsen).
- **NO diuretics.** Dobutamine is the preferred inotrope. Avoid positive-pressure ventilation unless essential.

## 2 APPROACH (ECHO-SELECTED)

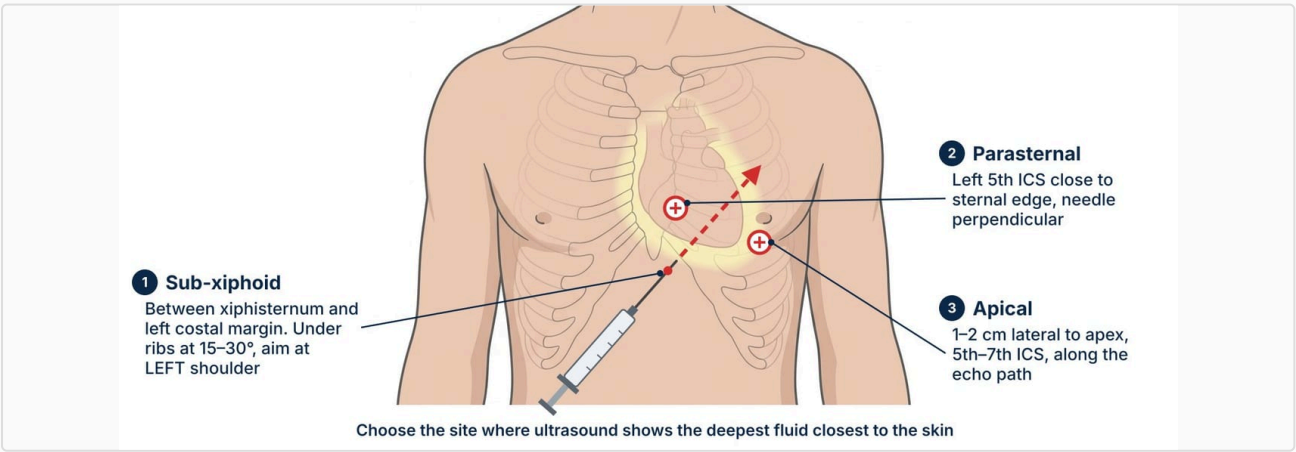
- Pause for 5 seconds. Confirm landmarks. Semi-reclining ~30°.
- **Apical:** 5th–7th ICS, 1–2cm lateral to apex, over rib. **Sub-Xiphoid:** 15–30° to LEFT shoulder. **Parasternal:** perpendicular, ≤1cm from sternal margin.

## 3 ENTER & CONFIRM

- Advance with aspiration. On fluid: advance ~2mm, slide cannula on, remove needle, pass J-wire (from CVC kit), dilate, railroad catheter.
- **Confirm:** 5ml agitated saline — bubbles in pericardial space.
- Bloody tap = myocardial puncture OR haemorrhagic effusion — confirm with agitated saline.

## 4 DRAIN & POST-PROCEDURE

- Remove enough to normalise BP/CVP — **not >1 L initially**; the rest over following hours (pericardial decompression syndrome).
- CXR. Drain: aspirate every 4–6h; remove when <25–30ml/24h. Urgent Cardiology/CT Surgery.




*Use Ultrasound to select best window*

*Feedback/Updates: jake.turner@uhb.nhs.uk*

Reference: ESC 2025 Guidelines for Myocarditis and Pericarditis | Technique: De Carlini & Maggiolini, ESC Cardiology Practice 2017.
Open Guideline

# EXTERNAL PACING

Page 1: Walkthrough (Preparation)

## INITIAL ACTIONS AND INDICATIONS CHECK

- Call 2222 "Medical Emergency".
- Call Cardiology urgently for temporary wire placement.
- **Life-threatening features:** shock, syncope, myocardial ischaemia, severe heart failure, immediately post-ROSC.
- **Drugs first:** Atropine `500mcg` IV, repeat to max `3mg` (glycopyrrolate alternative).
- **Risk of Asystole:** Recent asystole, Mobitz II, complete heart block with broad QRS, pause > 3s.

## EQUIPMENT

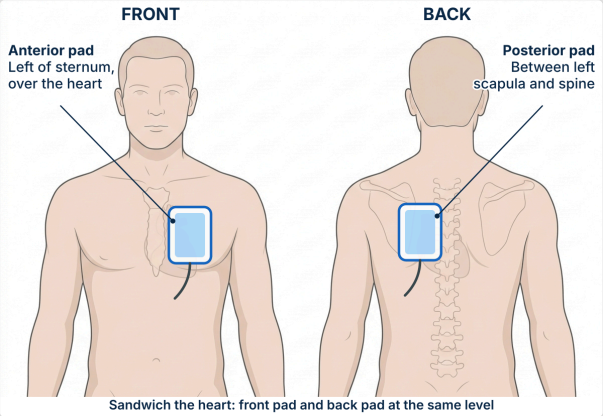
- [ ] Defibrillator (Pacer Mode).
- [ ] **Pacing Pads** (AP Position).
- [ ] **3-Lead ECG** (Required for sensing).
- [ ] Sedation (Ketamine/Midazolam). FETCH DRUG
- [ ] Analgesia (Fentanyl). FETCH DRUG

## POST-PROCEDURE PLAN

- Monitor conscious level (Sedation).
- Ensure cardiology review happens immediately.

## STEP-BY-STEP CONFIGURATION

1. **PADS:** Apply Anterior-Posterior (Sandwich the heart).
2. **LEADS:** Connect 3-LEAD ECG. Confirm R-wave markers on screen.
3. **MODE:** Select PACER. Mode: Demand.
4. **RATE:** Set to 70 bpm.
5. **CURRENT (mA):** Start `60 mA`; increase until capture (AAGBI QRH).
6. **ELECTRICAL CAPTURE:** Pacer Spike → Broad QRS → T wave.
7. **MECHANICAL CAPTURE:** Palpate Femoral Pulse. Measure blood pressure immediately after capture.
   **WARNING:** Skeletal muscle twitching is NOT capture. Carotid pulse is unreliable due to neck twitching.
8. **SETTING:** Find threshold. Set final output to Threshold + `10 mA`.




*Anterior-Posterior Position is Best*

# EXTERNAL PACING

**Page 2: Action (Unstable Bradycardia)**

- **Symptomatic Bradycardia unresponsive to drugs.**
- **Sedation:** Pacing is painful. Use Ketamine/Midaz.
- **Call:** 2222 Medical Emergency.

## 1 SETUP

- Pads: Anterior-Posterior preferred.
- Connect 3-Lead ECG (Critical for sensing).
- Select "Pacer". Rate: 70 bpm.

## 2 CAPTURE (MA)

- Start `60 mA`; increase until capture seen.
- **Electrical Capture:** Spike followed by Broad QRS.
- Ignore skeletal muscle twitching. Look at ECG.

## 3 CONFIRMATION & POST-PROCEDURE

- **Mechanical Capture:** Palpate Femoral Pulse. Measure BP immediately.
- Ensure pulse matches pacer rate.
- Set output `10 mA` above threshold.
- **After:** Refer to Cardiology for Transvenous/Permanent Pacemaker. Monitor conscious level.

**🛑 PLAN B: NO CAPTURE?**

- Check Pads/Leads. Swap Pad position.
- Isoprenaline `5 mcg/min` or Adrenaline `2–10 mcg/min` IV.
- Alternatives: aminophylline, dopamine, glucagon (β-blocker/CCB overdose).
- Consider Hyperkalaemia/Toxicity. Expert help → transvenous pacing.



*Feedback/Updates: jake.turner@uhb.nhs.uk*

Reference: Resus Council UK ALS 2025.
Open Guideline

# SUBCLAVIAN LINE

**Page 1: Walkthrough (Preparation)**

🧰 **HALO TROLLEY**

## INITIAL ACTIONS AND INDICATIONS CHECK

- Call 2222 only if in active cardiac arrest.
- Hypovolaemic Shock (Vein remains patent): use **large-bore 8.5 Fr introducer** for volume; triple lumen for drugs/pressors.
- C-Spine Immobilisation (IJ inaccessible).
- **Contraindications:** Clavicle/upper rib fracture. Coagulopathy (relative).
- Pneumothorax/chest drain on one side: go on THAT side.
- Head-down relatively contraindicated if raised ICP.

## EQUIPMENT

- ☐ Chlorhexidine.
- ☐ Central Line Kit (Triple Lumen) and/or 8.5 Fr introducer kit.
- ☐ Sterile Gown, Gloves, Drapes.
- ☐ Lidocaine 1%. FETCH DRUG
- ☐ Saline Flushes x3. FETCH SALINE
- ☐ Suture (Silk) + Scalpel.
- ☐ Dressings (Tegaderm).

## POST-PROCEDURE PLAN

- CXR Mandatory: exclude pneumothorax.
- Tip: lower SVC / cavo-atrial junction (~4cm below carina).
- **"Wire out, shout out"**: witness confirms, document.
- Secure well.

## LANDMARK TECHNIQUE

1. **POSITION:** 10–20° head-down. Arms by sides, head neutral. **No pillow/roll behind back** (narrows clavicle–1st rib gap). Cardiac monitor on.
2. **LANDMARK:** 1–2cm below junction of medial and middle thirds of clavicle, OR just below its midpoint.
3. **LOCAL:** Check allergies. Infiltrate skin and periosteum with SYRINGE.
4. **AIM:** Finger in Sternal Notch. Bevel up. Aim towards it, needle PARALLEL TO FLOOR. Walk under clavicle, aspirating constantly.
5. **WIRE:** Flash → rotate bevel + J-tip towards FEET. Advance 20–23cm; ectopics → withdraw until they stop. **Never let go of the wire.** Resistance on withdrawal: remove needle + wire together.
6. **CONFIRM VENOUS before dilating:** Dark, non-pulsatile blood is NOT proof in arrest/shock. USS along wire or manometry (IV tubing / transducer).
7. **DILATE** only until it just enters the vein → CATHETER to ~15cm (R) / 16–17cm (L).
8. **MAX 2–3 attempts.** New equipment each time.

# SUBCLAVIAN LINE

**Page 2: Action (Infraclavicular Approach)**

🧰 **HALO TROLLEY**

- **Ultrasound-guided if available** (Safe Vascular Access 2025). Landmark technique if no US.
- **Right side preferred (Lower pleural dome).**
- **Call: 2222 only if in active cardiac arrest.**

## 1 POSITION

- 10–20° head-down. Arms by sides. **Head neutral.** No roll behind back.
- If difficult: small roll under same-side shoulder, or 5cm caudal arm traction.

## 2 LANDMARKS

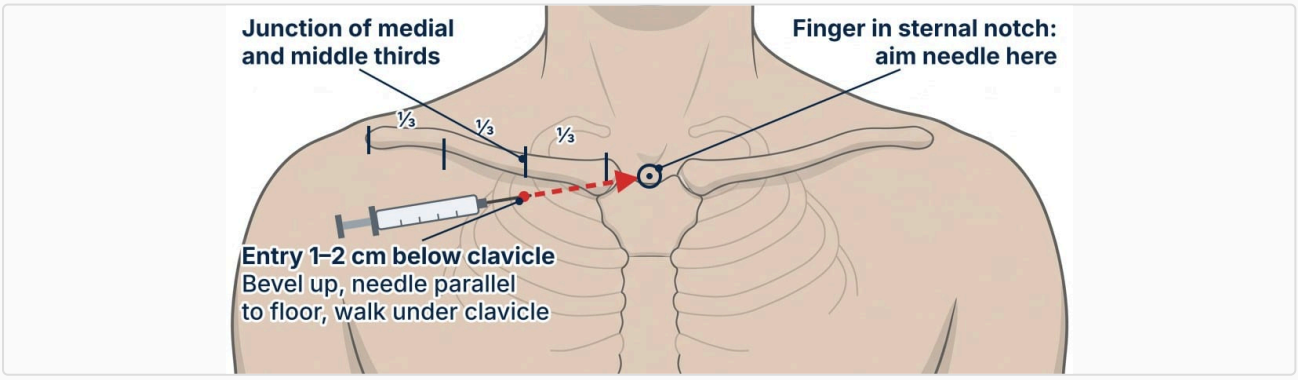
- **Time-out (LocSSIP).** Confirm landmarks.
- 1–2cm below medial/middle third junction (or just below midpoint). **Aim:** Sternal Notch.
- **Keep flat:** Parallel to floor (Avoids lung apex).

## 3 SELDINGER & POST-PROCEDURE

- Flash → bevel + J-tip to FEET → Wire 20–23cm (must slide easily; ectopics = pull back). **Never let go of wire.**
- **Confirm venous before dilating** (USS along wire / manometry). Dark blood is not proof in shock.
- Dilate just into vein → Catheter ~15cm R / 16–17cm L → Aspirate all ports → Suture.
- **"Wire out, shout out"** (witnessed). Max 2–3 attempts.
- **Artery:** needle → withdraw, press 10 min, watch for haemothorax. **Dilator/catheter in artery -> LEAVE IN**, call vascular/IR.
- **Mandatory CXR:** exclude pneumothorax; tip lower SVC / cavo-atrial junction.

**🛑 PLAN B: FAILED ACCESS?**

- Immediate Intraosseous (Humeral/Tibial) is the primary fallback for emergency access.
- Femoral Line.
- Ultrasound Guided Peripheral IV (Long catheter).




*Landmark: Medial/Mid 1/3 Clavicle → Sternal Notch*

*Feedback/Updates: jake.turner@uhb.nhs.uk*

Reference: Association of Anaesthetists Safe Vascular Access 2025; MSD Manual: Infraclavicular Subclavian Cannulation (Sep 2026).

Open Guideline

# LATERAL CANTHOTOMY

**Page 1: Walkthrough (Preparation)**

🧰 **HALO TROLLEY**

## INITIAL ACTIONS AND INDICATIONS CHECK

- Call Ophthalmology and MaxFax urgently. Call 2222 only if part of major trauma.
- **Retrobulbar Haematoma:** Orbital Compartment Syndrome.
- ⛔ **Suspected open/ruptured globe = do NOT perform.** Don't measure IOP or press on the globe; call ophthalmology.
- **Signs:** Proptosis, lids hard to open, rock hard eye, APD, vision loss, restricted eye movements, severe pain.
- **IOP:** > 40 mmHg.
- **No Tonometer?** Test by Palpation, Visual Acuity or Ocular US.
- **Ultrasound:** Look for "Guitar Pick Sign" (tenting of posterior globe).
- **Rule:** Do NOT wait for CT or ophthalmology. **Time is Vision:** damage may start within 12–15 min (older teaching ~100 min).

## EQUIPMENT

- ☐ Antiseptic (chlorhexidine/povidone-iodine): **skin only, must NOT enter eye** — or saline.
- ☐ **Sterile Drape / Gown Pack.**
- ☐ Lidocaine **1%** or **2%** with Adrenaline. FETCH DRUG
- ☐ 2ml Syringe + Orange Needle. Topical anaesthetic drops (proxymetacaine). FETCH DRUG
- ☐ Saline Irrigation.
- ☐ **Iris Scissors** + toothed forceps.
- ☐ **Artery Forceps** (small, e.g. mosquito).
- ☐ Sterile Gauze.
- ☐ Chloramphenicol eye ointment & Eye patch.

## POST-PROCEDURE PLAN

- Do NOT suture. Chloramphenicol eye ointment + eye patch.
- Urgent Ophthalmology Review.
- CT Orbit.
- Medical Management (Mannitol/Acetazolamide).

## PROCEDURE STEPS

1. **PREP:** Clean (saline / antiseptic kept out of eye). Anaesthetic drops. Check allergies. Inject 1–2ml LA with adrenaline into lateral canthus (aim away from globe). Uncooperative: sedate/restrain.
2. **CRUSH:** Clamp ARTERY FORCEPS horizontally across the lateral canthus (~ 1cm). Compress for 1 min (haemostasis).
3. **CANTHOTOMY (Skin):** Remove forceps. Cut the crushed line horizontally **1–2cm to the orbital rim** (≈ end of eyebrow) with SCISSORS. Point SCISSORS laterally and away from the eye to protect the globe.
4. **CANTHOLYSIS (Inferior Crus):**
   - Pull lower eyelid down and out.
   - Palpate the tight "guitar string" band (Inferior Crus).
   - "Strum" it with SCISSORS to identify.
   - **CUT** the Inferior Crus, aiming inferoposteriorly towards the rim.
5. **CHECK:** Lower lid should fall away. IOP should drop.
6. **ANY DOUBT?** Cut the **Superior Crus** too (upper lid band).

# LATERAL CANTHOTOMY

**Page 2: Action (Retrobulbar Haematoma)**

🧰 **HALO TROLLEY**

- **Proptosis + Trauma + Decreased VA / APD.**
- **IOP > 40 mmHg. Damage may start within 12–15 min — act now.** ⛔ **Open globe = do NOT perform.**
- **Call: Urgent Ophthalmology / Max Fax Review.**

## 1 PREPARE

- Pause for 5 seconds. Confirm landmarks.
- Check for severe allergies. Clean (keep antiseptic out of eye). Drops + LA (w/ Adrenaline).
- Protect globe during injection.

## 2 CRUSH & CUT

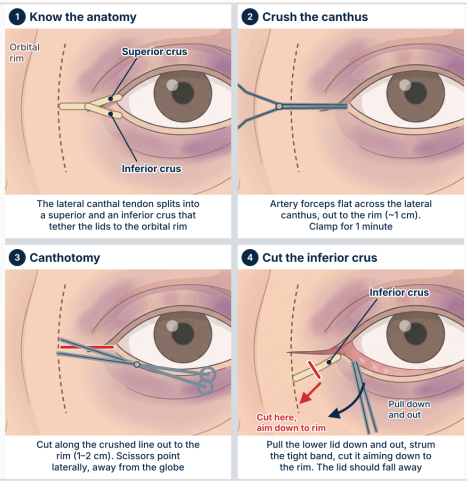
- Crush lateral canthus (1 min).
- Cut Skin & Orbicularis horizontally 1–2cm to orbital rim (Canthotomy). Point scissors laterally away from globe.

## 3 CANTHOLYSIS & POST-PROCEDURE

- Pull lower lid down. Identify **Inferior Crus** (Guitar String).
- **Cut Inferior Crus**, aiming inferoposteriorly.
- **Success:** Lower lid hangs loose. IOP drops.
- **After:** Do NOT suture. Chloramphenicol ointment + eye patch. Urgent Ophthalmology/MaxFax review. CT Orbit.

🛑 **PLAN B: ANY DOUBT / IOP STILL HIGH?**

- Cut Superior Crus.
- Ensure cuts are deep enough to rim.
- Medical Management (Mannitol/Acetazolamide).




*Canthotomy, then cut the inferior crus (cantholysis)*

*Feedback/Updates: jake.turner@uhb.nhs.uk*

Reference: RCEM Learning - Ocular Trauma & Sight for Sore Eyes | MSD Manual: Lateral Canthotomy (2025).
Open Guideline

# SEVERE MAX-FAX BLEED

Page 1: Walkthrough (Preparation)

🧰 HALO TROLLEY

## INITIAL ACTIONS AND INDICATIONS CHECK

- Call 2222 for "Major Haemorrhage" and "Anaesthetic Emergency".
- Call ENT Registrar and Anaesthetics immediately for airway control.
- **Airway:** Distortion, loose teeth, blood, swelling.
- **Plan A:** Conscious? Sit up + Suction.
- **Plan B:** RSI. Anticipate difficult view. Have eFONA ready.
- **Bleeding:** Maxillary artery is prolific. Scalp wounds bleed heavily.
- **C-Spine:** Spinal motion restriction, but do NOT prioritise over airway — no collar while a conscious patient needs to lean and spit. Once intubated the collar is also part of the midface splint (see right).

## EQUIPMENT

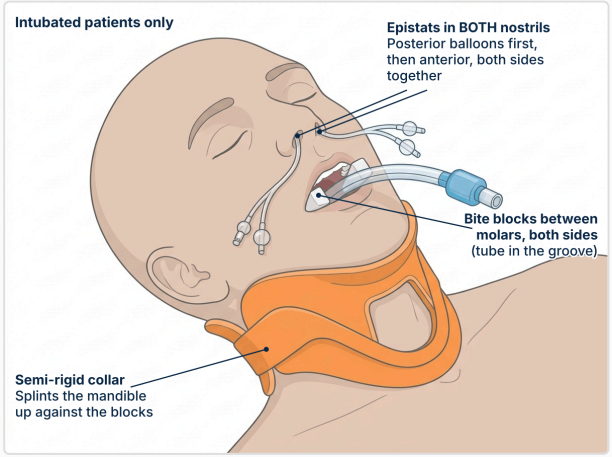
- [ ] Chlorhexidine.
- [ ] Suction x2 (Wide bore).
- [ ] **Bite Blocks** / McKesson Props (x2).
- [ ] Semi-rigid C-Spine Collar (sized) - mandible splint.
- [ ] **Option A:** Nasal Epistats (Double Balloon).
- [ ] **Option B:** TWO Foley Catheters (12-14F).
- [ ] Rapid Rhino Packs (7.5cm / 9cm).
- [ ] Heavy Suture (0 Silk) - Tongue Stitch.
- [ ] Dressings/Packs.
- [ ] TXA `1g` IV. FETCH DRUG

## POST-PROCEDURE PLAN

- CT Head/Face.
- Transfer Theatre/Interventional Radiology.

## HAEMORRHAGE CONTROL

1. **POSITION:** "Lean and Spit" if conscious. Suction constantly. Sweep airway with gloved finger or Magill forceps to clear avulsed teeth and bone fragments.
2. **MIDFACE REDUCTION (Le Fort):**
   - "Dish face" appearance.
   - Place gloved fingers behind upper teeth/palate.
   - PULL ANTERIORLY (Hard).
   - Disimpacts the maxilla. Stops bleeding.
   - Hold reduction; bite blocks go in once intubated (step 4).
3. **TONGUE STITCH:** Flail mandible? Stitch tongue tip and pull forward.
4. **PACK + SPLINT (INTUBATED ONLY, in this order):**
   **Option A: EPISTATS** (Double Balloon)
   a. Insert both sides along floor of nose — **do NOT inflate yet**.
   b. BITE BLOCKS between molars both sides (ETT in groove).
   c. Apply semi-rigid **C-SPINE COLLAR** — splints the mandible.
   d. Inflate POSTERIOR balloons both sides together (up to `10ml`) + gentle traction.
   e. Inflate ANTERIOR balloons both sides together (up to `30ml`).
   **Option B: FOLEYS + RAPID RHINO**
   - Insert BOTH FOLEY CATHETERS (one each side) posterior. Bite blocks + collar as b–c. Inflate posterior balloons with `10-15ml` of SALINE. Pull back.
   - THEN insert RAPID RHINOS Anteriorly.
   *Why: collar holds mandible → blocks hold palate against it → balloons fix maxilla. One-sided inflation can displace fractures.*




*Epistats + Bite Blocks + C-Spine Collar (midface splint)*

# SEVERE MAX-FAX BLEED

**Page 2: Action (Stabilisation)**

🧰 **HALO TROLLEY**

- **Conscious: Sit up, lean forward.**
- **Unconscious: RSI. Avoid nasal intubation.**
- **Call: 2222 Major Haemorrhage.**
- **Prepare for Front of Neck Airway (CICO risk high).**

## 1 MIDFACE REDUCTION

- **Indication:** Floating Maxilla (Le Fort).
- Sweep airway with gloved finger or Magill forceps to clear avulsed teeth and bone fragments.
- Gloved fingers behind upper teeth. **Pull Forward.**
- Hold reduction; bite blocks go in once intubated (step 2).

## 2 PACK + SPLINT (INTUBATED ONLY)

- **Order:** Epistats in (not inflated) → Bite blocks both sides → **C-Spine collar** → Inflate posterior balloons both sides together (up to `10ml`) + gentle traction → Inflate anterior both sides together (up to `30ml`).
- **Foley Method:** Insert Foleys **BEFORE** Rapid Rhinos.
- Inflate Foley (Posterior) with `10-15ml` of saline, pull back → Then Pack Anterior. Bite blocks + collar as above.
- Conscious patient leaning forward: no collar, no supine — spinal motion restriction never takes priority over the airway.

## 3 ADJUNCTS & POST-PROCEDURE

- **TXA** `1g` **IV** stat. Activate MHP. Transfuse early.
- Consider Tongue Suture if obstructing.
- **Transfer:** CT Head/Face. Theatre/Interventional Radiology.



*Feedback/Updates: jake.turner@uhb.nhs.uk*

Reference: RCEM Learning: Floating in the Face of Danger | GNAAS Maxillofacial Trauma SOP v4 (2022) | Harris et al. Eur J Emerg Med 2010;17:230-3.

Open Guideline

# LAST (LIPID RESCUE)

**Page 1: Walkthrough (Preparation)**

## INITIAL ACTIONS AND INDICATIONS CHECK

- Call 2222 "Anaesthetic Emergency".
- **History:** Recent LA injection (Lidocaine/Bupivacaine).
- **Prodrome:** Tongue tingling, metallic taste, tinnitus, agitation.
- **CNS:** Confusion → Seizures → Coma.
- **CVS:** Hypertension (Early) → Bradycardia → Broad QRS → Asystole.
- **Mimic:** Anaphylaxis (LAST usually has no rash/wheeze).

## LIPID RESCUE KIT

- [ ] **Intralipid 20%** (CHECK CONCENTRATION). FETCH DRUG
- [ ] Ensure minimum 1 Litre total stock of Intralipid is available.
- [ ] 50ml Syringe + Drawing needle.
- [ ] Wide Bore Cannulae x2.
- [ ] Giving Sets.
- [ ] Benzos (Midazolam). FETCH DRUG

## 📍 STORAGE LOCATION

**INTRALIPID IS KEPT IN THE FIRST CUPBOARD ON THE LEFT IN THE MAJORS DRUG CUPBOARD.**

## POST-PROCEDURE PLAN

- Consult TOXBASE and contact the National Poisons Information Service (NPIS).
- ICU Admission. Monitor for recurrence.
- Exclude pancreatitis: daily amylase or lipase. Report as critical incident.

## IMMEDIATE MANAGEMENT

1. **STOP INJECTING LA** (remember infusion pumps). Call for Help, arrest trolley & Lipid Kit.
2. **AIRWAY:** 100% O2; secure airway if needed. **Avoid hypercarbia — consider mild hyperventilation.**
3. **SEIZURES:** Small incremental **benzodiazepine = drug of choice**. Thiopental or propofol can be used — beware negative inotropy. Consider neuromuscular blockade if uncontrolled.
4. **CPR:** Standard CPR + lipid. **Adrenaline ≤1mcg/kg** (not 1mg). **Avoid vasopressin.** Avoid Lidocaine. Recovery may take >1 hour — consider cardiopulmonary bypass.
5. **LIPID BOLUS:** `1.5 ml/kg` (~ `100ml` for a 70kg adult). Give INTRALIPID **20%** over **2–3 min**. Repeat at 5 and 10 min if not stable.
6. **LIPID INFUSION:** `15 ml/kg/hr` (~ `1000ml/hr` for a 70kg adult). Double to 30 ml/kg/hr at any time after 5 min if not stable. Max cumulative 12 ml/kg.

# LAST (LIPID RESCUE)

**Page 2: Action (Local Anaesthetic Toxicity)**

- **CNS:** Tingling, metallic taste, seizure.
- **CVS:** Bradycardia, Broad QRS, Arrest.
- **Action:** STOP INJECTING LA (incl. pumps). Call for Lipid.
- **Call:** 2222 Anaesthetic Emergency.

## 1 AIRWAY & CPR

- 100% O2. Avoid hypercarbia — consider mild hyperventilation.
- Seizures: small incremental benzodiazepine (drug of choice).
- CPR: **Avoid Lidocaine**. Use Amiodarone.
- **Adrenaline ≤1mcg/kg** (not 1mg). Avoid vasopressin. Recovery may take >1h.

## 2 INTRALIPID 20% (BOLUS)

- **Bolus:** `1.5 ml/kg` (~ `100ml` for a 70kg adult).
- Give over **2–3 minutes**.
- Repeat bolus at **5 and 10 min** if not stable or circulation deteriorates.

## 3 INTRALIPID 20% (INFUSION) & AFTER

- **Rate:** `15 ml/kg/hour` (~ `1000ml/hr` for a 70kg adult).
- Double to 30 ml/kg/hr at any time after 5 min if not stable.
- Do not exceed 12 ml/kg cumulative (70kg: 840ml).
- **Post-Procedure:** ICU. Monitor for recurrence. Daily amylase/lipase (pancreatitis).



*Feedback/Updates: jake.turner@uhb.nhs.uk*

Reference: Association of Anaesthetists Quick Reference Handbook - Local Anaesthetic Toxicity (2023).
Open Guideline

# SURGICAL AIRWAY (EFONA)

**Page 1: Walkthrough (Preparation)**

🔴 **DIFFICULT AIRWAY TROLLEY**

## EQUIPMENT LOCATED IN DIFFICULT AIRWAY TROLLEY

### INITIAL ACTIONS AND INDICATIONS CHECK

- Call 2222 "Anaesthetic and Airway Emergency".
- Declare "CAN'T INTUBATE, CAN'T OXYGENATE".
- ONE final upper-airway oxygenation attempt if not done. **Give neuromuscular blocking drug now.**

### EQUIPMENT

- ☐ **Scalpel (Size 10).**
- ☐ Bougie (Coude tip).
- ☐ ETT Size 6.0 (Cuffed).
- ☐ Lubricant.
- ☐ 10ml Syringe (Cuff).
- ☐ Tube Tie.
- ☐ BVM / EtCO2.

### POST-PROCEDURE PLAN

- Secure tube.
- Sedate.
- Transfer to ENT Theatre / ICU.

### DAS PLAN D (SCALPEL-BOUGIE-TUBE)

1. **POSITION:** Neck Extended. Pillow under shoulders.
2. **HANDSHAKE:** Stabilise larynx with non-dominant hand. Palpate CTM with Index.
3. **INCISION:** DAS 2025 default = **vertical midline skin incision** up to 8cm (caudad → cephalad). Blunt finger dissection to CTM. Then transverse stab through CTM, blade edge **TOWARDS YOU**. (Transverse stab alone only if CTM clearly palpable or pre-marked.)
4. **ROTATE:** Rotate SCALPEL 90 degrees (Sharp edge towards feet).
   **CRITICAL:** DO NOT WITHDRAW BLADE.
5. **BOUGIE:** Insert BOUGIE down the flat side of blade. Advance 10 to 15cm. Feel for "clicks" or "hold up".
6. **RAILROAD:** Remove SCALPEL. Railroad lubricated 6.0 ETT over bougie. Twist to pass membrane.
7. **VENTILATE:** Inflate cuff. Connect BVM. **Confirm ETCO2**. Secure.

# SURGICAL AIRWAY (EFONA)

**Page 2: Action (CICO Rescue)**

🔴 **DIFFICULT AIRWAY TROLLEY**

⚠️ **DECLARE "CAN'T INTUBATE, CAN'T OXYGENATE". OPEN AIRWAY TROLLEY. CALL 2222 ANAESTHETIC AND AIRWAY EMERGENCY.**

## 1 DECLARE & PREPARE

- **Call out:** "CAN'T INTUBATE, CAN'T OXYGENATE. Proceeding to Front of Neck."
- **Action:** 100% O2. **Give neuromuscular blocker.** Extend Neck maximally.
- Identify CTM (Laryngeal Handshake).

## 2 INCISION (SCALPEL)

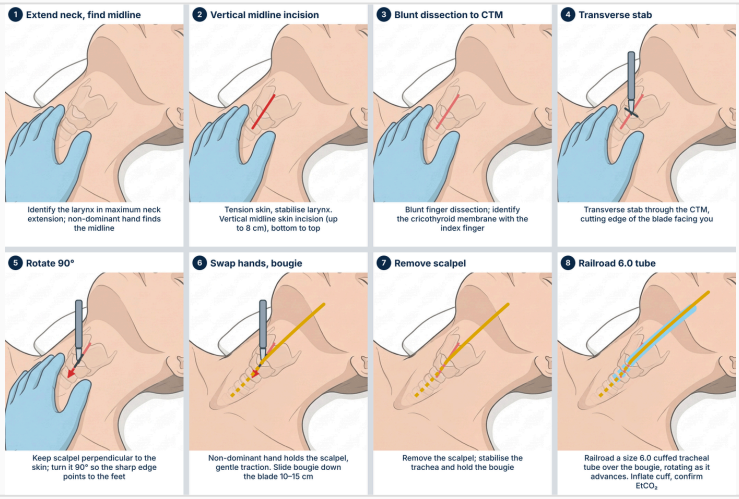
- Pause for 5 seconds. Confirm landmarks.
- **Vertical skin incision** up to 8cm → finger dissection → **transverse stab** through CTM.
- **Turn Blade 90 degrees** (Sharp edge caudal/towards feet).
- **DO NOT REMOVE SCALPEL.** Swap hands if needed.

## 3 RAILROAD (BOUGIE-TUBE)

- Slide **Bougie** down blade into trachea.
- Remove Scalpel. Railroad **6.0 ETT**.
- Twist tube to pass membrane. Inflate cuff.

## 4 CONFIRMATION & POST-PROCEDURE

- Connect BVM. **Confirm EtCO2 (Gold Standard).**
- Suction trachea. Secure tube. Sedate.
- **After:** Transfer to ENT Theatre / ICU.




*DAS 2025 scalpel-bougie-tube, step by step*

*Feedback/Updates: jake.turner@uhb.nhs.uk*

Reference: Difficult Airway Society (DAS) 2025 Guidelines.
Open Guideline

# STOP 5 HOT DEBRIEF

Perform immediately after the event, before team disperses.

1. **S**ummarise the case (1 minute).
2. **T**hings that went well?
3. **O**pportunities to improve?
4. **P**oints for action (Fix equipment, reporting).
5. **5** minutes max.

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Is everyone okay? Does anyone need specific support right now?
*Check on your colleagues.*